# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The 10th term of $21, 18, 15, 12, \ldots$ is: 1

**(a)** -6 **(b)** -3 **(c)** -9 **(d)** -30

**2.** The 5th term from the end of the AP with last term 100 and $d = 4$ is: 1

**(a)** 116 **(b)** 80 **(c)** 88 **(d)** 84

**3.** If $a_n = 3n + 2$, then the common difference is: 1

**(a)** 3 **(b)** 2 **(c)** 5 **(d)** 6

**4.** If $S_n = n^2 + 2n$, then $a_1 =$ 1

**(a)** 2 **(b)** 4 **(c)** 3 **(d)** 1

**5.** The sum of first 6 positive odd numbers is: 1

**(a)** 72 **(b)** 36 **(c)** 42 **(d)** 30

**6.** The 10th term of the AP $\frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ is: 1

**(a)** 6 **(b)** $\frac{9}{2}$ **(c)** 5 **(d)** $\frac{11}{2}$

**7.** The 15th term of $8, 13, 18, 23, \ldots$ is: 1

**(a)** 78 **(b)** 83 **(c)** 75 **(d)** 73

**8.** The 8th term from the end of the AP with last term 80 and $d = 3$ is: 1

**(a)** 56 **(b)** 59 **(c)** 62 **(d)** 101

**9.** If $a_n = 3n + 5$, then the common difference is: 1

**(a)** 8 **(b)** 5 **(c)** 6 **(d)** 3

**10.** If $S_n = n^2 + 3n$, then $a_1 =$ 1

**(a)** 4 **(b)** 5 **(c)** 3 **(d)** 1

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Which term of the AP $3, 8, 13, 18, \ldots$ is 63? 2

**12.** Find the sum of the first 10 multiples of 3. 2

**13.** Find the sum of the AP $-5, -2, \ldots, 28$. 2

**14.** Which term of the AP $8, 11, 14, 17, \ldots$ is 71? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** In an AP, the 2nd term is 7 and the 5th term is 19. Find the AP and its 10th term. **3**

**16.** How many terms are there in the AP $18, \frac{31}{2}, 13, \ldots, -\frac{19}{2}$? **3**

**17.** Find the sum of the first 22 terms of the AP $15, 19, 23, 27, \ldots$ **3**

**OR**

In an AP, the 3rd term is 11 and the 7th term is 23. Find the AP and its 12th term. **3**

**18.** How many terms are there in the AP $6, 10, 14, \ldots, 86$? **3**

**OR**

Find the sum of the first 25 terms of the AP $-9, -7, -5, -3, \ldots$ **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** How many terms of the AP $24, 22, 20, 18, \ldots$ must be taken so that the sum is 144? **5**

**OR**

A loan is repaid in 20 monthly instalments; the first is ₹100 and each later instalment is ₹10 more than the previous. Find the instalment in the last month and the total amount repaid. **5**

**20.** How many terms of the AP $9, 17, 25, 33, \ldots$ must be taken so that the sum is 450? **5**

**OR**

A loan is repaid in 12 monthly instalments; the first is ₹200 and each later instalment is ₹15 more than the previous. Find the instalment in the last month and the total amount repaid. **5**

## END OF QUESTION PAPER — Total: 40 Marks

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